

AutoValuate Intelligence

AI-powered used-car valuation for the UAE — free, browser-based, deployed

Executive Brief · 3 pages

SP Jain School of Global Management — Student Venture

19 July 2026

AutoValuate Intelligence is a **live, working web application** that tells a used-car buyer or seller in the UAE what a car is really worth — in seconds, for free. It combines two independent AI systems: a **computer-vision model** that inspects uploaded photos for physical damage, and a **machine-learning pricing model** trained on real UAE marketplace listings. The entire product runs on free-tier cloud infrastructure and is built and operated by a student team.

15.65%

median pricing error — 46%
better than a naive make/model
baseline

1,302

real UAE listings used to train
the pricing model

78.9%

of true prices fall inside our
stated confidence range (target
80%)

AED 0

cost to the user — no paid tiers,
no sign-up required

01 What the product does

- **Instant price estimate** — enter a car's details, get a price range with a stated confidence level, not a single guessed number.
- **On-device damage scan** — upload photos; an AI inspects them for dents, scratches, cracks and more. Photos never leave the user's device.
- **Explainable pricing** — every estimate shows exactly which factors (mileage, age, brand, damage) moved the price and by how much.
- **Market comparables** — the estimate is shown against real, similar listings currently on the market.
- **Dealer bulk valuation** — a dealer can upload a spreadsheet of an entire fleet and get every car valued at once.
- **Open, free API** — any developer can call the pricing engine directly; no account or key required.

02 How the pieces fit together



All four stages run on free-tier cloud services — no paid infrastructure.

03 Technology stack, in plain terms

LAYER	WHAT WE USE	WHAT IT'S FOR
Website	Next.js (React) on Vercel	The app users see and interact with in their browser
Damage scan	YOLOv8 object-detection model, run in-browser	Finds and labels visible car damage in a photo, on the user's own device
Price engine	XGBoost (gradient-boosted trees)	Predicts fair market price from a car's specs and condition
Backend	FastAPI (Python) on Render, with an AI agent orchestrator	Combines the price engine, the damage scan and market data into one report
Database	Supabase (Postgres + vector search)	Stores comparable listings and shareable report links

04 How the two AI engines work

Engine 1 — Damage Detection (Computer Vision)

A model called **YOLOv8** looks at each uploaded photo and draws a box around anything that looks like damage — a dent, a scratch, a crack, broken glass, a broken light, a flat tire, a puncture, or a missing part (8 categories in total). It converts what it sees into a single condition score out of 100. This runs entirely on the user's own phone or laptop, so no photo is ever sent to our servers — a genuine privacy advantage, not just a technical choice.

Engine 2 — Price Estimation (Machine Learning)

A model called **XGBoost** was trained on 1,302 real UAE car listings. Given the make, model, age, mileage, specification and condition, it predicts a fair price — and, importantly, a **range** around that price with a stated confidence level, rather than pretending to know an exact number.

The Orchestrator

A layer of coordinating logic takes the output of both engines, pulls in comparable listings, and assembles one explainable report — the price, why it is that price, and how confident the system is.

05 Key terms explained

YOLOv8 / ONNX

YOLOv8 is the damage-detection AI. ONNX is the portable file format it's exported to so it can run inside a web browser instead of on a server.

XGBoost

A widely-used, highly accurate machine-learning technique for predicting a number (here, price) from structured data like mileage and age.

mAP (mean Average Precision)

The standard score for "how good is the damage detector" — the industry way to grade computer-vision accuracy.

APE / Median APE

"Absolute Percentage Error" — how far off a single price guess was, in percent. We report the median across all test cars, which is more honest than an average.

Confidence range (split-conformal prediction)

A statistically calibrated price range — e.g. "80% confident the true price falls in this band" — rather than one number presented as certain.

Agentic orchestration

AI "agents" that each handle one job (pricing, damage, market lookup, write-up) and hand results to each other automatically to build the final report.

06 Results — verified, not marketing

METRIC	RESULT	WHAT IT MEANS
Pricing accuracy (median error)	15.65%	Typical price estimate is within ~16% of the actual sale price
Improvement vs. naive baseline	+46.1%	Roughly twice as accurate as simply averaging similar listings
Confidence-range reliability	78.9%	Very close to the 80% target — the stated ranges can be trusted
Damage-detector accuracy (mAP)	0.732	Strong on close-up damage photos; see honesty note below
Browser ↔ server scoring match	56 / 56	The score shown to the user always matches what the backend independently computes

Read this the way a CFO would want it read: the 0.732 detector score was measured on a validation set that overlaps with training data and covers only 6 of 8 damage types — it is not a number we would defend to an external auditor. The pricing numbers above, by contrast, are measured on a genuinely held-out test set and are safe to repeat externally.

07 Known limitations — stated plainly

- 1 The damage scan is sensitive to how a photo is framed.**

The same damage, photographed at a slightly different crop or distance, can score up to 47 points differently. We have diagnosed the root cause precisely (the detector was originally trained on close-up damage photos, not whole-car shots) and scoped the fix — retraining with wide-shot images — but have not yet been able to run it, since it requires more real UAE car photos and GPU training time we don't yet have.
- 2 Pricing accuracy is limited by data volume, not by the model.**

1,302 training listings caps typical accuracy at roughly 15–16% error. Published industry benchmarks assume a dataset around 15 times larger. More listings — not more tuning — is the lever that moves this number.
- 3 A licensing question blocks closed-source commercial resale.**

The damage detector is derived from an open-source model licensed AGPL-3.0. That license permits free use exactly as we operate today, but would require either a paid commercial license or a replacement model before this could be sold as closed-source software.
- 4 There is currently no revenue.**

The product is intentionally 100% free with no paid tiers — a deliberate choice while the data and licensing questions above remain open, not an oversight.
- 5 Coverage is UAE-only.**

All training data, comparable listings and market assumptions are specific to the UAE market today.

08 What's next

- | | | |
|---|--|---|
| 1
Retrain the damage detector
Run the already-written retraining process on a larger, wide-shot photo set to fix the framing sensitivity above. | 2
Grow the listings dataset
Expand past 1,302 UAE listings — the single biggest lever on pricing accuracy. | 3
Resolve the licensing path
Decide between an enterprise license or a licence-compatible replacement model before any commercial launch. |
|---|--|---|

09 Project context & approach

Who built this

A student team at SP Jain School of Global Management, built and operated as a combined academic and venture-track project — not a funded startup.

Why free-tier only

Every service in the stack (hosting, database, model inference) runs on a free pricing plan by design. It proves the product works before any cash is spent, and keeps the cost of serving a user at effectively zero.

How claims in this document are checked

Every number above is read from a committed evaluation report, re-runnable from the source repository — nothing here is an estimate or a rounded-up marketing figure.

How limitations are found

Weak spots (like the framing sensitivity on page 3) are found by deliberately trying to break the system before a user does, not discovered after launch.

Live product: auto-valuate-intelligence.vercel.app
API: autovaluate-api.onrender.com

Source & full evaluation reports: github.com/krish2105/AutoValuate-Intelligence
License: AGPL-3.0 · built and operated on free-tier cloud services only